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ORIGINAL RESEARCH

Supplementation of Probiotics and Its Effects on Physically Active Individuals and Athletes: Systematic Review

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The aim of this study was to conduct a systematic review of the effects of probiotic supplementation on physically active individuals. The participants, interventions, comparisons, outcome and study design inclusion criteria were (a) studies involving healthy adults or older subjects of both sexes who did physical exercise (including athletes and physically active individuals), (b) interventions with probiotics, (c) inclusion of a control group, (d) outcomes not previously defined, and (e) clinical trials and randomized clinical trials, with no language or date restrictions. The search was conducted in the following scientific databases: MEDLINE, Embase, SciELO, Scopus, and Lilacs. Search terms were "Probiotics" OR "Prebiotics" OR "Microbiota" AND "Exercise" OR "Athletes." The articles were first screened by title and abstract by two independent reviewers and disagreements resolved by a third reviewer. Data were extracted independently by the same two reviewers; results were extracted in duplicate and then compared to avoid errors. A total of 544 articles were retrieved and 24 were included. A total of 1,680 patients were included, most of them being male ($n = 1{,}134$, 67.5%), with a mean age of 30.9 ± 6.1 years. Following probiotic supplementation, positive effects have been reported for several outcomes including respiratory tract infection, immunologic markers, and gastrointestinal symptoms in both athletes and nonathletes. However, published studies have distinct protocols and measured outcomes, and some of them have small sample size and failed to prove beneficial effect on probiotic supplementation, leading to inconclusive results for standardized supplementation protocols.

***Keywords**:* dietary supplements, exercise, microbiota, respiratory tract infection

Probiotics are live microorganisms, usually lactic acid bacteria, that when administered in adequate amounts, bring health benefits to the host (Rijkers et al., 2011). Their usage has widely increased, mainly as dietary supplements in capsules, powder or fermented milk (Pyne et al., 2015). *Lactobacillus*, *Bifidobacterium*, and *Saccharomyces* are some bacterial genera that are actively used as probiotics (Azad et al., 2018; Kerry et al., 2018).

Several beneficial effects have been attributed to probiotics. They were previously linked to antipathogenic, antidiabetic, anti-obesity, anti-inflammatory, anticancer, antiallergic, and angiogenic activities, along with their action on the brain and central nervous system (Kerry et al., 2018). Probiotics act by producing antipathogenic compounds, stimulating defensins production and competing for pathogen binding and receptor sites. They are also involved in the production of short chain fatty acids, synthesis of neuroactive compounds and immunomodulators, such as interleukin-10, interleukin-12, and interferon-γ (Kerry et al., 2018). Moreover, probiotics have been proposed to modulate gut microbiota composition by increasing favorable bacteria populations, improve gut epithelium barrier function, and favor cytokine production (Azad et al., 2018).

Gut microbiota plays a crucial role in the development of the innate and adaptive immune systems in humans (De Palma et al., 2014), by modulating gut motility, intestinal barrier homeostasis, and energy metabolism (Hooper et al., 2012). A healthy intestinal microbiota is stable and acts in the metabolism of toxic products of barely digestible polysaccharides, serving as a barrier against pathogens, and aiding in the development of the host immune system (Tanaka & Nakayama, 2017). This symbiotic environment is associated to other healthy conditions, as well as dysbiosis, defined as an alteration and imbalance of gut microbiota, and is associated with metabolic disorders (Cani, 2018). Both intestinal and extraintestinal benefits have been reported by probiotic supplementation including irritable bowel syndrome (Salem et al., 2018), obesity, Type II diabetes mellitus (Kerry et al., 2018), and airways diseases (Martens et al., 2018).

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Probiotics could provide beneficial effects on the gut microbiome for physically active individuals, since high-intensity exercises can negatively affect the gut barrier function by decreasing splanchnic blood flow by up to 80% of baseline levels and increasing gut permeability (Briza et al., 2010; Monda et al., 2017). Intense exercise triggers an inflammatory and immune response, increases reactive oxygen species and unbalance of the brain–gut axis, and also increases body temperature and mechanical damage (Clark & Mach, 2016). On the other hand, low-intensity endurance training plays a protective role in decreasing intestinal transit time and consequently the gastrointestinal exposure to pathogens (Briza et al., 2010; Monda et al., 2017).

Physical exercise and dietary adaptation are linked with changes in the composition of the gut microbiome regardless of probiotic use. It was previously reported that athletes have differences in fecal microbiota when compared with sedentary controls (Barton et al., 2017). The use of probiotics among physically active individuals could become a strategy to further improve intestinal health, respiratory symptoms, and physical performance (Bermon et al., 2015; Ibrahim et al., 2017; Komano et al., 2018; Monda et al., 2017), given their proposed beneficial effects. However, their efficacy in this population remains unproven. Therefore, the aim of this study was to provide a systematic review of clinical trials evaluating the effect of probiotic supplementation in a physically active population.

Methods

Eligibility Criteria and Search Strategy

This systematic review was designed following the preferred reporting items for systematic reviews and meta-analyses recommendations and registered in the international prospective register of systematic reviews (number CRD42017070151).

Inclusion criteria were based on the participants, interventions, comparisons, outcome and study design framework: (a) studies involving healthy adults or older subjects of both sexes who did physical exercise (including athletes and physically active), (b) interventions with probiotics, (c) inclusion of a control group, (d) outcomes not previously defined (as open question; all outcomes evaluated by included studies were reported), and (e) clinical trials and randomized clinical trials, with no language or date restrictions.

The search for articles was conducted in MEDLINE, Embase, SciELO, Scopus, and Lilacs databases between February 2017 and March 2017 using the terms—"Probiotics" OR "Prebiotics" OR "Microbiota" AND "Exercise" OR "Athletes." The search was updated in February 2018. Full search strategy is described in Supplementary Table S1 (available online).

Data Extraction and Analysis of the Methodological Quality of the Studies

After duplicate articles were removed, articles were first independently screened by title and abstract by two of the authors (G.B. Möller and M.J.V. da Cunha Goulart) and disagreements were resolved by a third reviewer (F.D. Alves). After, articles selected in this first stage were read for identification of inclusion and exclusion criteria and data extraction, also performed by two independent reviewers (G.B. Möller and M.J.V. da Cunha Goulart), extracted in duplicate and compared to avoid errors. From each article, the data extracted were author(s) and year, age and sex, exercise, intervention and control, main outcome, and results.

Cohen's kappa coefficient was calculated between the studies selected by the reviewers. Methodological quality of the studies was analyzed according to the Grading of Recommendations Assessment, Development, and Evaluation criteria.

Results

A total of 544 articles were retrieved in the initial search; 140 duplicate articles were removed and 374 articles excluded after screening by title and abstracts. The 30 remaining articles were read in full; eight of them excluded for not meeting the inclusion criteria—three articles because of the study population and four articles because of the study design. In the update of the revision, two more articles were added, totaling 24 included. We could not contact two authors, which were excluded. Twenty-four articles were included in the qualitative analysis (Figure 1). Agreement between reviewers was 92% and $\kappa = .836$; $p < .001$. Two articles were included in the search performed in February 2018.

Quality of the Articles

The quality of evidence was assessed by the Grading of Recommendations Assessment, Development, and Evaluation approach. Most of the studies were randomized clinical trials, and, hence, initially classified as "high quality." Two reviewers worked through each of the Grading of Recommendations Assessment, Development, and Evaluation criteria (risk of bias, indirectness of evidence, inconsistency of evidence, imprecision of effect size, and publication bias; Supplementary Table S2 [available online]).

Participants and Characteristics of the Studies

A total of 1,680 patients were included, most male (approximately 67.5%, $n = 1{,}134$), and mean age of 30.9 ± 6.1 years. All studies were written in English, published between 2007 and 2018, and included trained subjects regardless of ethnicity and the type/intensity of physical exercises.

Only two of the 24 articles did not report the type of physical exercise. In the other studies, the types of exercises varied between sports, physical training, and physical tests.

Characteristics of the Interventions

Supplementation with *Lactobacillus* was the most common intervention, being used in 21 studies, followed by *Bifidobacterium*, presented in 10 studies. The supplementation with probiotics containing strains from the *Lactobacillus* genus used in the studies included the species *acidophilus*, *fermentum*, *casei*, *gasseri*, *rhamnosus*, *brevis*, *bactis*, *helveticus*, *paracasei*, *plantarum*, *bulgaricus*, *casei Shirota*, *salivarius*, and *lactis* in dose varying from 3×10^8 to 1×10^{11} colony-forming units. Probiotics containing strains from the *Bifidobacterium* genus included species as *bifidum*, *logum*, *lactis*, *breve*, and *animalis ssp. lactis*. *Enterococcus faecium*, *Streptococcus thermophilus*, *Bacillus coagulans,* kefir, and *raftilose* GR were also reported among the probiotic interventions.

Main Results

The studies assessing athletes and/or physically active individuals are presented in Table 1.

Respiratory tract outcomes. With regard to respiratory tract symptoms, five articles showed a positive effect of probiotics (Clancy et al., 2006; Cox et al., 2008; Marinkovic et al., 2016; Moreira et al., 2007; Strasser et al., 2016; West et al., 2011),

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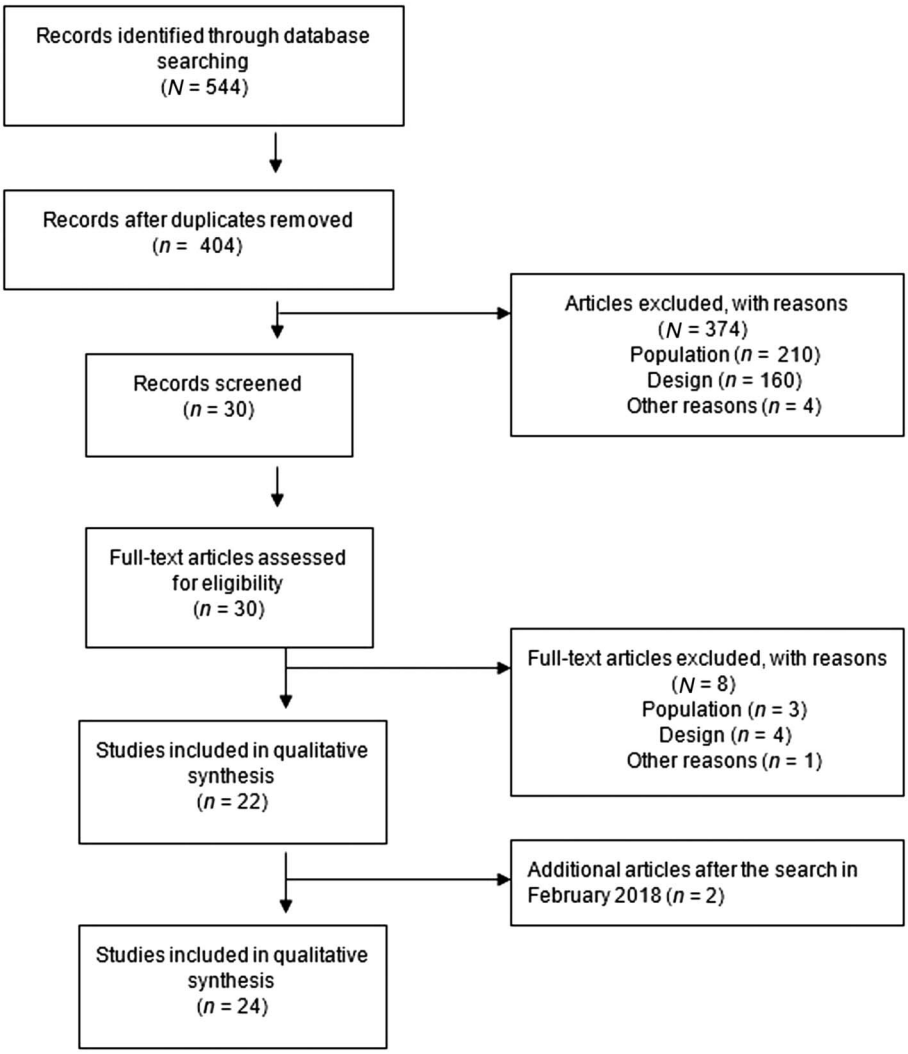

Figure 1 — Flow chart of study selection.

assessing different outcomes. Cox et al. (2008) and Marinkovic et al. (2016) reported lower duration of respiratory tract illness. The lower frequency of symptoms was found by Strasser et al. (2016) and Marinkovic et al. (2016). Reduction on severity of symptoms was observed by West et al. (2011), but not by Marinkovic et al. (2016). West et al. (2014) reported a lower risk of upper respiratory illness episode. Gleeson et al. (2011) found lower frequency of subjects with respiratory symptoms and number of respiratory episodes, but no significant results regarding severity and duration of symptoms. The intervention with probiotics showed no effect on respiratory tract in three studies (Haywood et al., 2014; Kekkonen et al., 2007; Moreira et al., 2007).

Gastrointestinal outcomes. About the gastrointestinal symptoms, three studies showed positive, but not entirely clear results (Gleeson et al., 2011; Kekkonen et al., 2007; West et al., 2011). Kekkonen et al. (2007) found lower duration of gastrointestinal symptom episodes, but not incidence; while West et al. (2011) reported a reduction in the severity of gastrointestinal symptoms at higher training loads in men, but possibly increase in gastrointestinal symptoms in well trained female athletes. Gleeson et al. (2011) demonstrated beneficial effects of probiotics on days with gastrointestinal discomfort, but not in symptom prevalence. Two studies showed no difference on gastrointestinal outcomes (Haywood et al., 2014; Strasser et al., 2016). The study of Haywood et al. (2014) found significant differences on respiratory and gastrointestinal distress when all symptoms were analyzed together, and there was no evaluation of exercise performance.

Regarding the gastrointestinal permeability, Lamprecht et al. (2012) found a decrease in zonulin in stool, whereas Shing et al. (2013) failed to demonstrate changes in other parameters of gut permeability. Although the study of West et al. (2012) reported that composition of microbiota changes with probiotics, the gut permeability assessment did not differ between intervention and control groups (West et al., 2012), while preserved gastrointestinal permeability was observed with probiotics in another study (Roberts et al., 2016).

Immunity or metabolic outcomes. One study (Abbasi et al., 2015) evaluating bodybuilding athletes found positive effects of the intervention with probiotics after acute aerobic exercise protocol (without performance evaluation). The intervention group decreased total cholesterol and triglycerides, but showed no difference on insulin, glucose, and high-density lipoprotein-cholesterol levels.

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Table 1 Summary of the Studies With Probiotics on Athletes and/or Physically Active Individuals

Population, *N*	Author (year)	Intervention/control	Exercise	Main outcomes	Results
Body builders (3 days/week, regular exercise at least 5 years) $N = 14$—M Age 20–55 years	Abbasi et al. (2015)	INT: *Lactobacillus casei* ($10^9 \times 5.1$ CFU/g), *Lactobacillus acidophilus* ($10^9 \times 2$ CFU/g), Lactobacillus C. ($10^9 \times 5.1$ CFU/g), *Lactobacillus bulgaricus* ($10^8 \times 2$ CFU/g), *Bifidobacterium breve* ($10^{10} \times 2$ CFU/g), *Bifidobacterium longum* ($10^9 \times 7$ CFU/g), *Streptococcus thermophilus* ($10^9 \times 5.1$ CFU/g) capsule supplementation for 28 days CONT: placebo (starch)	Acute aerobic test—Bruce protocol—performed before and after supplementation	Improvement in glycemic (glucose and insulin) and lipid profile (total cholesterol, triglycerides, and HDL-cholesterol)	Total cholesterol and triglyceride levels lower in INT group with exercise vs. CONT group (170.5 vs. 196.5 mg/dl; $p = .0008$; and 154.5 vs. 252.2 mg/dl). No effects on insulin, glucose, and HDL-cholesterol levels
Recreational athletes $N = 27$—17 M/10 F 18 healthy athletes (median age 26.1 years) and nine fatigued athletes (median age 24.7 years; self-referred)	Clancy et al. (2006)	INT: *L. acidophilus* (2×10^{10} CFU) capsule for 4 weeks CONT: none	Not reported	Inflammation (whole blood culture secretion of IL-4, IFN-γ, and IL-12; salivary IgA)	The whole blood culture secretion of IFN-γ increased (7.8–32.1 pg/ml; $p = .01$) in fatigued athletes to levels found in healthy athletes. The changes did not differ between the healthy and fatigued athlete groups
Highly trained distance runners $N = 20$—M Mean age 27.3 ± 6.4 years	Cox et al. (2008)	INT: *Lactobacillus fermentum* (1.2×10^{10} CFU) supplementation capsules followed by a *washout* of 28 days with *L. fermentum* VRI-003 (1.2×10^{10} CFU) capsules for 28 days CONT: placebo (microcrystalline cellulose)	Usual training	Performance (treadmill running test to assess mileage, duration, intensity, and VO_2max); illness (episodes of sore throat, cough, runny nose, headache, fever, and others) mucosal and systemic immunity (salivary IgA, whole blood cultures of IL-4, IL-12, and IFN-γ); and respiratory tract (infection or illnesses)	Lower symptom days of respiratory tract illnesses (30 vs. 70; $p < .001$). No effect on treadmill performance or immunity parameters (salivary IgA, whole blood cultures of IL-4, IL-12, and IFN-γ)
Endurance-trained runners $N = 8$—M Mean age 26 ± 6 years	Gill et al. (2016)	INT: probiotic beverage containing *L. casei* (10^{11} CFU) for 7 days followed by a *washout* of 1 month CONT: placebo (similar but without probiotic)	2-hr running exercise at 60% VO_2max in hot ambient conditions (34 °C and 32% relative humidity)	Salivary antimicrobial proteins (IgA, α-amylase, lysozyme, and cortisol)	Besides an increment in salivary IgA and α-amylase concentration and secretion rates immediately after exercise ($p < .05$), there were no differences between groups. There was no effect on salivary lysozyme and cortisol. There was no evaluation of performance
Elite rugby n = 30—M Mean age 24.7 ± 3.6 years	Haywood et al. (2014)	INT: *Lactobacillus gasseri* (2.6 billion CFU), *Bifidobacterium bifidum* (0.2 billion organisms), and *B. longum* (0.2 billion organisms) supplementation capsule for 4 weeks followed by a *washout* of 4 weeks CONT: placebo (cornflour)	4 days/week and 4 hr/day—1 day of competition and 2 days of recovery and light training. Training included gym work, weights, skills, and fitness tests	Symptoms of upper respiratory tract, gastrointestinal, and other infections	The number of players who reported an incidence for any symptom was significantly higher on the placebo ($n = 24$; 80%) compared with the probiotic ($n = 16$; 53%) supplementation ($p = .033$). There were no differences for respiratory tract or gastrointestinal distress exclusively

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Table 1 *(continued)*

Population, *N*	Author (year)	Intervention/control	Exercise	Main outcomes	Results
Marathon runners (personal-best marathon time <3 hr 45 min for women and <3 hr 30 min for men) $N = 141$—125 M/16 F Median age 40 years	Kekkonen et al. (2007)	INT: *Lactobacillus rhamnosus* GG (4×10^{10} CFU) in two 65-ml bottles of milk-based fruit drink for 3 months. In total, 27 subjects ($n = 70$) received capsules (1×10^{10} CFU) CONT: placebo (similar but without probiotic). In total, 28 subjects ($n = 71$) received placebo capsules	Supplementation for 3 months before marathon race	Number of healthy days, incidence of upper respiratory tract infections, and gastrointestinal symptoms	The duration of gastrointestinal-symptom episodes in the probiotic and placebo groups was 1.0 vs. 2.3 days, respectively, during the 2 weeks after the marathon ($p = .046$). No significant effect in other outcomes during the 3-month training period or 2 weeks after the marathon was found
Triathletes, runners, cyclists ($VO_2max > 45$ ml·kg^{-1}·min^{-1}) $N = 23$—M Age 30–45 years	Lamprecht et al. (2012)	INT: multispecies probiotics (*B. bifidum*, *Bifidobacterium lactis*, *Enterococcus faecium*, *L. acidophilus*, *Lactobacillus brevis*, and *Lactobacillus lactis*—dose of 2.5×10^9 CFU/g) in powder sachets (4 g/day) for 14 weeks ($n = 11$) CONT: placebo (cornstarch, maltodextrin, vegetable protein, $MgSO_4$, $MnSO_4$, and KCl; $n = 12$)	Incremental cycle ergometer test at 80 rpm. After a 3-min rest phase sitting inactive on the ergometer, work rate started at 60 W for 3 min and was increased 20 W every minute until voluntary exhaustion	Gastrointestinal permeability (zonulin and α1-antitrypsin); oxidation (carbonyl groups on proteins, malondialdehyde, and total oxidation status); and inflammation (TNF-α and IL-6)	Zonulin decreased ~20% in probiotic group compared with placebo after supplementation ($p = .019$). TNF-α showed reduced concentrations compared with the placebo group but this effect barely failed significance ($p = .054$). There was no effect of supplementation on α1-antitrypsin, carbonyl groups on proteins, malondialdehyde, total oxidation status, and IL-6
Elite athletes (training >11 hr/week) $N = 39$—29 M/10 F Age 18–28 years	Marinkovic et al. (2016)	INT: *Lactobacillus helveticus* Lafti (2×10^{10} CFU) capsule for 14 weeks ($n = 20$) CONT: placebo (maltodextrin; $n = 19$)	Incremental treadmill ergometer test	Incidence of respiratory infection; symptoms and shortening of illness duration; immunity and inflammatory parameters (leukocyte count, TGF-β, IL-10, and IFN-γ); and sport performance	Lower duration ($p = .047$), number of symptoms per episodes ($p = .035$), and number of infected days ($p < .001$) of respiratory events. There were no differences in immunity and inflammatory parameters and exercise performance
Amateur athletes $N = 24$—M Mean age 32.03 ± 6.12 years	Martarelli et al. (2011)	INT: *L. rhamnosus* and *Lactobacillus paracasei* (10^9 CFU) capsule for 4 weeks ($n = 12$) CONT: none ($n = 12$)	Maintenance of usual cycling training (prolonged intense exercise program)	Oxidative stress (reactive oxygen metabolites and biological antioxidant potential)	Less increment in reactive oxygen metabolites ($p < .05$) and higher biological antioxidant potential after exercise in probiotic group ($p < .01$), indicating high antioxidant activity during exercise
Ultramarathon runners $N = 32$—26 M/6 F Age 23–53 years	Marshall et al. (2017)	INT: *L. acidophilus* (10 billion CFU), *B. bifidum* (9.5 billion CFU), *B. lactis* (0.5 billion CFU), fructooligosaccharides (55.8 mg) capsule alone ($n = 11$), and *L. acidophilus* (2 billion CFU), *B. bifidum* (50 billion CFU), *B. lactis* (0.95 billion CFU), *Lactobacillus salivarus* (5 billion CFU) with L-glutamine (0.9 g) powder ($n = 10$) for 12 weeks CONT: none ($n = 11$)	294-km marathon	Analysis of Hsp72 and VO_2max	No difference between the groups in Hsp72 or VO_2max
Marathon runners (best marathon time <3 hr 40 min for women and <3 hr 30 min for men) $N = 139$—123 M/16 F Mean age 30 ± 9 years	Moreira et al. (2007)	INT: *L. rhamnosus* GG (3×10^8 CFU/ml) in two 65-ml bottles of milk-based fruit drink for 3 months. ($n = 70$); 27 subjects received capsules (5×10^9 CFU) CONT: placebo (similar but without probiotic). In total, 28 subjects ($n = 69$) received placebo capsules	Usual exercises (3 months and 1 week before marathon) before and after the marathon	Allergic inflammatory markers (eosinophil cationic protein, blood eosinophils, and total IgE)	No effect on allergy-related parameters between groups

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Table 1 *(continued)*

Population, *N*	Author (year)	Intervention/control	Exercise	Main outcomes	Results
Trained runners (mean VO_2max 62.2 ± 2.1 ml·kg^{-1}·min^{-1}) *N* = 10—M Mean age 27 ± 2 years	Shing et al. (2013)	INT: *L. acidophilus* (7.4 billion CFU), *L. rhamnosus* (15.55 billion CFU), *L. casei* (9.45 billion CFU), *Lactobacillus plantarum* (3.15 billion), *L. fermentum* (1.35 billion CFU), *B. lactis* (4.05 billion CFU), *B. breve* (1.35 billion CFU), *B. bifidum* (0.45 billion CFU), and *Streptococcus thermophiles* (2.25 billion) capsule for 4 weeks followed by a *washout* of 3 weeks CONT: placebo (skim milk powder)	Running to fatigue at 80% of their ventilatory threshold at 35 °C and 40% humidity	Gastrointestinal permeability (urinary claudin 3, lactulose:rhamnose ratio in urine after consumption of 5-g lactulose, 5-g rhamnose, and 5-g sucrose and serum lipopolysaccharide); systemic inflammation (IL-1ra, TNF-α, IL-6, IL-10, and neutrophil elastase production); and running performance (time–fatigue run)	Increased the time to fatigue (min:s INT 37:44 ± 2:42 vs. CONT 33:00 ± 2:27; $p = .033$). No differences in gastrointestinal permeability and systemic inflammation
Trained athletes (cardiorespiratory response and fitness of ≥150% of reference values during maximal exercise) *N* = 29—13 M/16 F Age 20–35 years	Strasser et al. (2016)	INT: *B. bifidum*, *B. lactis*, *E. faecium*, *L. acidophilus*, *L. brevis*, and *L. lactis* (1×10^{10} CFU) sachets for 12 weeks ($n = 14$) CONT: placebo sachets (similar without probiotics; $n = 15$)	Maintenance of endurance training. Cycle ergometer exercise test until exhaustion	Respiratory infections, illness symptoms, gastrointestinal discomfort, serum tryptophan, phenylalanine, kynurenine and tyrosine, and neopterin (immune activation marker), performance (VO_2max)	Lower frequency of respiratory infection symptoms in probiotic group (35% vs. 79%; $p = .016$). Reduced exercise-induced tryptophan degradation rates in the probiotic group ($p < .05$). No differences in other parameters evaluated
Competitive cyclists *N* = 99—64 M/35 F Mean age 27 ± 2 years	West et al. (2011)	INT: *L. fermentum* (1 billion CFU; PCC®—ProBio PCC; Pharmanex, Provo Utah) capsule for 11 weeks ($n = 47$) CONT: placebo (microcrystalline cellulose; $n = 50$)	Maintenance of usual cycling training	Gastrointestinal and respiratory symptoms; mucosal immunity (lactoferrin, lysozyme, and salivary IgA); and systemic immunity (IL-1ra, IL-6, IL-8, IL-10, TNF-α, and IFN-γ)	Reduction in the severity of self-reported symptoms and illness load of lower respiratory illness (0.31; 99% CI [0.07, 0.96]), use of cold and flu medication (0.57; 99% CI [0.01, 3.1]), and severity of gastrointestinal symptoms at higher training loads (0.7; 99% CI [0.2, 1.2]) in male athletes. No differences in immunity parameters
N = 84—54 M/30 F Age 18–55 years (mixed population: recreational to competitive athletes)	Gleeson et al. (2011)	INT: 65 ml of fermented milk containing *L. casei* Shirota (6.5×10^9 CFU) for 16 weeks ($n = 32$) CONT: placebo (similar without probiotic; $n = 26$)	Maintenance of usual training (endurance) Exercise: 10 hr/week	Illness and gastrointestinal symptoms, immune and inflammatory profile (lymphocytes, IFN-γ, TNF-α, IL-1β, IL-2, IL-4, IL-6, IL-8, IL-10, macrophage chemoattractant protein-1, and salivary IgA)	Subjects with respiratory symptoms for one or more weeks (90% vs. 66%; $p = .021$), number of respiratory episodes (2.1 ± 1.2 vs. 1.2 ± 1.0; $p < .01$), and proportion of days with gastrointestinal discomfort (3% vs. 2%; $p = .008$) were higher in CONT than INT group. Salivary IgA was higher in INT than CONT group ($p < .03$). There was no difference in immune profile
N = 54—M/F Age 18–35 years (mixed population: recreational to competitive athletes)	Gleeson et al. (2012)	INT: *Lactobacillus salivarius* (2×10^{10} CFU) powder sachet for 4 months ($n = 27$) CONT: placebo (maltodextrin; $n = 27$)	Maintenance of usual training (endurance) Exercise: 9 hr/week	Upper respiratory tract infection, salivary antimicrobial proteins (IgA and lysozyme), and blood white cells (lymphocytes)	No significant effects

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Table 1 *(continued)*

Population, *N*	Author (year)	Intervention/control	Exercise	Main outcomes	Results
$N = 268$—155 M/113 F Age 18–32 years (mixed population: recreational to competitive athletes)	Gleeson et al. (2016)	INT: *Lactobacillus casei* (6.5×10^9 CFU) 65 ml of fermented milk twice a day for 20 weeks ($n = 137$) CONT: placebo drink ($n = 131$)	Maintenance of usual training Endurance $n = 66$ Individual sports $n = 33$ Team games $n = 169$	Upper respiratory tract infection episode incidence, duration and severity, and plasma cytomegalovirus and Epstein–Barr antibodies	There was no difference in upper respiratory tract infection episode, duration, and severity Plasma cytomegalovirus and Epstein–Barr antibodies reduced in INT group compared with placebo ($p < .001$) in seropositive participants
$N = 21$—M Age 19–26 years (exercise no more than twice per week)	Ibrahim et al. (2017)	INT: combination of multistrain probiotic (*L. acidophilus, L. casei, L. lactis, B. bifidum, Bifidobacterium infantis, B. longum*, with 3×10^{10} CFU, twice a day) sachets powder ingestion for 12 weeks ($n = 9$) CONT: placebo (similar powder without probiotics; $n = 12$)	Circuit training three times per week with two circuits of exercises from Weeks 1 to 8 followed by three circuits of exercises from Weeks 9 to 12	Muscular strength and power (peak torque), IL-6, and IL-10	There was no difference between groups that exercised (circuit training) regarding probiotic supplementation
$N = 29$—M Mean age 21.5 ± 2.8 years (at least 3 months of recreational training)	Jäger et al. (2016b)	INT: combination of *Bacillus coagulans* (1 billion CFU) and protein (20 g casein) for 14 days after a *washout* of 7 days (prior protein supplementation only) CONT: placebo with casein	Single-leg leg press (10×10; 60 s), leg extension (5×12; 60 s), and rear foot elevated split squat (5×12; 60 s)	Muscle thickness (ultrasonography), soreness and recovery (visual analogue scales), participants' strength (one-repetition maximum testing), and power (cycle ergometry)	Significantly increased 24 and 72 hr recovery and decreased muscle soreness 72 hr after training ($p < .05$)
$N = 15$—M Mean age 25 ± 4 years (resistance training for 1 year prior)	Jäger et al. (2016a)	INT: 3 weeks before and after exercise with a 21-day *washout* period receiving *B. breve*(5 billion CFU) and *S. thermophiles* (5 billion CFU) capsule CONT: placebo	Isometric exercises (eccentric exercise)	Isometric strength, muscle soreness, range of motion and girth, and blood IL-6 and creatine kinase	Probiotic decreased IL-6 (–43%; 90% CI [–91.3, –7.8]), enhanced isometric average peak torque production at 24–72 hr following exercise ($p < .05$). No effects of probiotics on muscle soreness, range of motion and girth, and creatine kinase
$N = 67$—M/F Age 18–24 years	O'Brien et al. (2015)	INT: Kefir (in full-fat cow milk; 10^9–10^{10} CFU) for 15 weeks; 2×/week ($n = 13$ in exercise training and $n = 21$ active control) CONT: placebo (control beverage; $n = 10$ in exercise training and $n = 21$ active control)	Exercise training subgroup: supervised long-distance running (endurance training). Test of 1.5-mile walk/run in the fastest possible time Active control subgroup: usual workout routine without supervision	Performance (1.5-mile time), body fat percentage, C-reactive protein, and consumer acceptance	Improvement (average of 4.11%) in 1.5-mile time in both exercise training subgroups. Attenuation of exercise-induced inflammation, since C-reactive protein increases only in the control group of exercise training. No differences in body composition. No effects on active controls. Good consumer acceptance for all groups
$N = 20$—18 M/2 F Mean age 35 ± 1 years (training for first long distance triathlon)	Roberts et al. (2016)	INT: *L. acidophilus* (10 billion CFU), *B. bifidum* (9.5 billion CFU), *B. lactis* (0.5 billion CFU), and fructooligosaccharides (55.8 mg) for 12 weeks before and 6 days after running ($n = 10$) CONT: placebo (cornflour; $n = 10$)	Triathlon training program designed to achieve a minimum of 80% of daily total training volume set	Plasma endotoxin units (limulus amebocyte lysate chromogenic quantification) and gastrointestinal permeability (5-hr urinary lactulose:mannitol recovery), race times in triathlon	Endotoxin units reduced 6 days postrace ($p < .01$) and gastrointestinal permeability remained unchanged with probiotics. No effects on race times

(continued)

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Table 1 *(continued)*

Population, *N*	Author (year)	Intervention/control	Exercise	Main outcomes	Results
$N = 22$—M Mean age 33.9 ± 6.5 years (cycling, regular physical activity as $VO_2max > 45$ ml·kg^{-1}·min^{-1})	West et al. (2012)	INT: glycomax immunoglobulin (200 mg), glycomax lactoferrin (50 mg), and CBARBlend-100 (Chr. Hansen A/S, Horsholm, Denmark) (100 mg): *L. paracasei* (4.6×10^8 CFU), *Bifidobacterium animalis ssp. lactis* (6×10^8 CFU), *L. acidophilus* LA-5 (4.6×10^8 CFU), *L. rhamnosus* GG (4.6×10^8 CFU), Raftiline (90 mg), Raftilose GR (10 mg), and magnesium stearate (10 mg) capsules for 21 days ($n = 11$) CONT: prebiotic acacia powder (116 mg), microcrystalline cellulose (23 mg), silica colloidal anhydrous (8 mg), chocolate flavor (31 mg), calcium hydrogen phosphate (174 mg), 116.5 lecithin epikuron, and magnesium stearate (5 mg; $n = 11$)	Usual exercise with registration of duration (km/week) and intensity (in a 1–5 scale: 1, *easy*; 5, *maximum*)	Mucosal immunity (salivary lactoferrin and albumin), systemic inflammation (IL-16, IL-18, IL-12, and IFN-γ), gut permeability (urinary lactulose and mannitol), fecal short chain fatty acids, and microbial analysis	Symbiotic supplementation (INT) significantly altered the composition of the gut microflora compared with prebiotic supplementation ($p < .001$), with increased *L. paracasei* numbers (ninefold 95% CI [1.2, 64]; $p = .03$). Symbiotic was associated with a 50% (95% CI [−12, 72]; $p = .02$) smaller increase in IL-16. No effect of probiotics on mucosal immunity, gut permeability, and other parameters
$N = 465$—240 M/224 F Age 18–60 years (three or more session of physical activity a week)	West et al. (2014)	INT: *B. animalis lactis* (2×10^9 CFU; $n = 161$) or *L. acidophilus* and *B. animalis lactis* (5×10^9 CFU each) sachet for 150 days followed by 14 days washout ($n = 155$) CONT: placebo ($n = 149$)	Maintenance of usual exercise	Episodes of respiratory and gastrointestinal illness, cold medication usage, and doctor visits	The risk of an upper respiratory illness episode was significantly lower in the *B. animalis lactis* group (hazard ratio 0.73; 95% CI [0.55, 0.95]; $p = .022$) compared with placebo There was a delay in the median time to an illness episode in the INT groups (3.2 and 3.4 months) compared with placebo (2.5 months). There were insufficient GI illness episodes for analysis. There was no difference in cold medication usage or doctor visits

Note. INT = intervention; CONT = control; M = male; F = female; VO_2max = maximal oxygen consumption; IL = interleukin; IFN = interferon; TGF = transforming growth factor; CI = confidence interval; TNF = tumor necrosis factor; CFU = colony-forming units; HDL = high-density lipoprotein; IgA = immunoglobulin A.

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Most studies did not find the effects of probiotic supplementation on immunity in athletes and/or physically active individuals (Cox et al., 2008; Gill et al., 2016; Gleeson et al., 2012; Marinkovic et al., 2016; Moreira et al., 2007; West et al., 2011; West et al., 2012).

In a mixed population of recreational to competitive athletes, one study (Gleeson et al., 2011) observed higher salivary immunoglobulin A in the intervention group, and another (Gleeson et al., 2016) found reduced cytomegalovirus and Epstein–Barr virus antibodies after 20 weeks of probiotic supplementation in seropositive individuals.

Considering inflammation, five studies did not find improvement on inflammatory markers (Gleeson et al., 2011; Ibrahim et al., 2017; Lamprecht et al., 2012; Marinkovic et al., 2016; Shing et al., 2013). Positive results were observed in four studies (Clancy et al., 2006; Jäger et al., 2016b; O'Brien et al., 2015; West et al., 2012). Reduction in interleukin-6 (Jäger et al., 2016b) and C-reactive protein (O'Brien et al., 2015) was also reported. West et al. (2012) demonstrated a smaller increase in interleukin-16 after probiotic supplementation, but no difference in interleukine-12, interleukine-18, and interferon-γ. The study of Clancy et al. (2006) reported that fatigued athletes have benefits with interferon-γ with probiotic supplementation.

Considering the antioxidant activity of probiotics in athletes, one study (Martarelli et al., 2011) found this benefit and one study (Lamprecht et al., 2012) did not.

Performance outcomes. The performance outcome was verified in seven studies, but evidence is weak. Shing et al. (2013) observed a difference in time to exhaustion between groups after 4 weeks of probiotics supplementation, but this cannot be attributed to the intervention because no measurement was performed before the supplementation. Strasser et al. (2016) found reduced exercise-induced tryptophan degradation but the relevance of this to performance is not known. Performance (evaluated by VO_2max on a cycle ergometer) remained unchanged over time, and this pattern did not differ between the probiotics and placebo groups. Jäger et al. (2016a) reported positive results on muscle recovery and soreness, but not strength; while Jäger et al. (2016b) observed enhanced peak torque production, but no difference in soreness, range of motion, and girth. Another five studies (Cox et al., 2008; Ibrahim et al., 2017; Marinkovic et al., 2016; Marshall et al., 2017; O'Brien et al., 2015)) failed to demonstrate the association between probiotic supplementation and exercise performance. Although not the primary objective of the study, Lamprecht et al. (2012) also found no difference in the performance of endurance trained men submitted to a cycle ergometer test.

Discussion

This systematic review, about the effects of probiotic supplementation in trained subjects, showed improvements in respiratory tract infections, immunological markers, and gastrointestinal symptoms but little evidence of benefits to exercise performance. However, the interventions have distinct protocols, regarding probiotic dose and strains, outcomes assessed, and other variables.

The practice of regular exercise is a well-established strategy to improve inflammatory conditions (O'Sullivan et al., 2015), especially when compared with a sedentary lifestyle. In studies evaluating the impact of exercise on gut microbiota, it was shown that moderate training causes changes in microbiota in humans, which varied according to body composition and exercise maintenance (Allen et al., 2018). Low-intensity exercise can increase the abundance of health-promoting bacteria (Bressa et al., 2017), and low–moderate exercise may improve immune function (Cook et al., 2015), oxidative stress, and inflammation (Sallam & Laher, 2016), with beneficial consequences to intestinal health. In physically active subjects, some studies included in this review it was demonstrated that probiotics provide further benefits to these parameters (Jäger et al., 2016b; O'Brien et al., 2015; Roberts et al., 2016; West et al., 2012). One mechanism by why probiotics may give rise to their benefits is through improving the microbiota, which was demonstrated by West et al. (2012), by providing an increase on *L. paracasei* amount in the microbiota of trained individuals.

The high-intensity, extenuating exercise, per se, in the absence of a nutritional or dietary intervention, can negatively affect the microbiota, local immune function, and microbiota brain axis, by a mechanism not fully elucidated (Hamasaki, 2017). Also, high-intensity exercise, which is part of athletes' training, involves the synthesis of metabolites and inflammatory mediators, and biological adaptations to exercise may lead to a decrease in proinflammatory cytokines, suggesting a regulatory interaction between exercise biology and host immunity (O'Sullivan et al., 2015). The impact of probiotics on metabolic parameters in athletes is not elucidated. Most studies did not find positive effects on immunity (Cox et al., 2008; Gill et al., 2016; Marinkovic et al., 2016; Moreira et al., 2007; West et al., 2011) or inflammation (Lamprecht et al., 2012; Marinkovic et al., 2016; Shing et al., 2013), except one that observed that fatigued athletes have benefits for interferon-γ with probiotic supplementation (Clancy et al., 2006). The antioxidant activity of probiotics in athletes is also unclear.

Athletes have exhaustive training, and there is a direct relationship between health status and performance. We showed that probiotics could improve aspects related to a better health status; however, there is no clear consensus. The respiratory and gastrointestinal symptoms seem to improve with probiotic supplementation, but not all studies agree (Gleeson et al., 2012; Haywood et al., 2014). These studies vary as to the composition of probiotics, doses, product presentation, time of evaluation, outcomes assessment, and population. Different studies found distinct results depending on the outcome assessed, for this reason it is not possible to identify a study characteristic that contributes to positive or absent results.

The probiotic benefit on performance and strength is also an unclear issue. Some studies found conflicting results (Jäger et al., 2016a, 2016b; Strasser et al., 2016), whereas other found difference in time to exhaustion (Shing et al., 2013), or none (Cox et al., 2008; Ibrahim et al., 2017; Marinkovic et al., 2016; Marshall et al., 2017; O'Brien et al., 2015). The study that observed impact on probiotics showed differences in time to exhaustion between groups (Shing et al., 2013); however, the comparison was performed between intervention and control groups only after exercise, without previous measurement. Therefore, the impact of probiotics on performance needs to be further elucidated.

The studies that evaluated mechanisms of action for probiotics; however, also exhibit divergent results. Some authors demonstrated positive effects of supplementation on inflammation, such as the C-reactive protein (O'Brien et al., 2015). The impact on interferon-γ, interleukin-6, and other cytokines are controversial (Clancy et al., 2006; Gleeson et al., 2011; Ibrahim et al., 2018; Jäger et al., 2016b; Lamprecht et al., 2012; Marinkovic et al., 2016; Shing et al., 2013; West et al., 2012). The same applies to immunity parameters, with substantial studies that did not find improvement

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associated with probiotic supplementation (Cox et al., 2008; Gill et al., 2016; Gleeson et al., 2012; Marinkovic et al., 2016; Moreira et al., 2007; West et al., 2011; West et al., 2012), except for the study of Gleeson et al. (2011) that reported higher salivary immunoglobulin A in intervention group. On the other hand, the beneficial effect on gut permeability seems to exist, as evidenced by studies of 12–14 weeks intervention (Lamprecht et al., 2012; Roberts et al., 2016). One shorter study of 4 weeks (Shing et al., 2013), however, did not report this finding. Again, most studies assessing inflammatory and immune markers presented different protocols, supplementation time, strains, and sample population, which impairs grouping characteristics to find the best intervention.

Although the relationship between diet, intestinal microbiota, host immunity, and metabolism are more evident in recent literature, the relationship between microbiota and exercise has not been fully explored and needs further research. Differences between the studies may be due to different types of exercise, training intensity, and type and dosage of probiotic therapy (West et al., 2012). One limitation that should be considered in this analysis is the wide variety of bacterial strains among the interventions, making it difficult to make associations between therapy and outcomes.

Despite the known restoring effects of exercise on the gut microbiota, it is too early to assign an adjunct role to exercise and probiotics. There are still challenging questions that require new tools and methodologies. These efforts should ultimately contribute to the knowledge of host–microbial interactions and provide new insights into health promotion and performance improvement strategies in athletes and physically active individuals.

Conclusions

The evidence indicates that probiotic supplementation can lead to positive effects on disease markers, including respiratory tract infection, immunologic markers, and gastrointestinal symptoms in physically active population. However, published studies have distinct protocols and measured outcomes, and some of them have small sample sizes, and failed to prove beneficial effect on probiotic supplementation.

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